

transaction costs of investors.¹⁸⁵ As discussed below, the Commission has conducted analysis to show that quoted and effective spreads are likely to decline such that costs of executing small and medium trades will likely decline.¹⁸⁶ Further, Rule 612, as amended, while simplified compared to the proposal, continues to be designed to address constraint concerns with respect to those NMS stocks. Market participants and investors will be able to more easily adapt to the amended tick regime because they will only need to accommodate, and adjust for, one additional minimum pricing increment that is already familiar for a limited, readily discernable, group of NMS stocks.¹⁸⁷ The \$0.005 minimum pricing increment for quotes and orders, one of the three additional ticks proposed by the Commission, was widely supported by commenters.¹⁸⁸ Price improvement on exchanges and ATs often occurs through midpoint executions in an increment of \$0.005. Accordingly, \$0.005 is an appropriate increment to introduce smaller, sub-penny minimum pricing increments in the national market system for quotes and orders priced equal to or greater than \$1.00.

Some individual commenters did not support the proposal.¹⁸⁹ One of those commenters stated that the minimum pricing increment for quotes and orders should be “based solely on that which can be spent in real life; no less than a single penny.”¹⁹⁰ Preexisting Rule 612 allowed quotes and orders in NMS stocks priced less than \$1.00 per share to be accepted, ranked and

¹⁸⁵ See *infra* section VII.D.1.b.ii.

¹⁸⁶ See *infra* section VII.D.1.b.ii.

¹⁸⁷ See *infra* section VII.D.1.a.

¹⁸⁸ See, e.g., MEMX Letter at 15-16. See also note 219 and accompanying text.

¹⁸⁹ See, e.g., letters from Joshua Russell dated Dec. 27, 2022; Matthew Gayvin Mutman dated Mar. 7, 2023; Aswin Joy dated Mar. 7, 2023.

¹⁹⁰ See Letter from Joshua Russell dated Dec. 27, 2022. But see letter from Anonymous dated Apr. 1, 2023 (stating “[g]etting more precise increment should be easy enough with our modern computers. At the gas station I get charged down to the .000th place, so why shouldn’t our markets work the same? Seems fair to me.”).

displayed in an increment as small as \$0.0001. Similarly, certain RLP Programs for national securities exchanges have been granted Commission exemptions to permit quotes and orders in NMS stocks priced equal to, or greater than, \$1.00 per share to be accepted, ranked and displayed in an increment as small as \$0.001. Sub-penny increments also existed in the market for many years, even prior to the adoption of Rule 612 in 2005.¹⁹¹ Sub-penny increments can allow market participants to better convey prices at which they are willing to trade, which can promote better price competition and lead to better price discovery. Further, as discussed above, sub-penny trading occurs frequently, whether at the midpoint or in other sub-penny increments.¹⁹² Thus, sub-penny increments are not a novel concept. As discussed above, \$0.005 is a common trading increment because of the use of midpoint orders under current Rule 612, and the ability to use such orders will not change under amended Rule 612. Nonetheless, the Commission understands that market participants may decide to provide investor notice and education about the availability of the new increment.¹⁹³

Another commenter stated that the proposed variable minimum pricing increments were “not an effective solution to address concerns related to tick-constrained stocks” and suggested a

¹⁹¹ Prior to decimalization, quotes and orders were made in increments that were fractions of a dollar, including 1/8, 1/16 and 1/32, which resulted in sub-penny pricing.

¹⁹² See supra section III.A.

¹⁹³ One commenter stated that to the extent the minimum quoting increment is reduced, FINRA would need to update the Manning Rule (FINRA rule 5320 which protects customer limit orders by requiring a minimum amount of price improvement for a firm to execute an order on a proprietary basis while holding an unexecuted customer limit order—the minimum amount of price improvement is currently \$0.01 for orders equal to or greater than \$1) in an equivalent manner. See Citadel Letter I at 8. The compliance date of the adopted rule provides sufficient time for FINRA to determine whether it would want to amend the Manning Rule in light of the amendments to Rule 612 and to file a proposed rule change pursuant to Section 19(b) of the Exchange Act and rule 19b-4 thereunder.

uniform \$0.001 minimum pricing increment for all NMS stocks.¹⁹⁴ A uniform \$0.001 minimum pricing increment for all NMS stocks goes beyond what is necessary to address the issues related to NMS stocks that are currently constrained by the \$0.01 tick. A \$0.001 minimum pricing increment would be significantly smaller than the current uniform \$0.01 minimum pricing increment for quotes and orders for NMS stocks that are priced equal to, or greater than, \$1.00 per share. A sub-penny increment for NMS stocks that is too small would increase the incidence of stepping ahead (i.e., pennyning)¹⁹⁵ and costs would not justify the benefits.

2. Specific Comments on the Proposed Minimum Pricing Increments

A few commenters did not support the implementation of the smallest proposed sub-penny increments (i.e., \$0.002 and \$0.001), and referenced certain concerns, including stepping ahead of displayed orders, quote flickering that occurs when the price of a trading center's best displayed quotations changes multiple times in a single second, and decreased depth.¹⁹⁶ Each of these were articulated as concerns by the Commission when Rule 612 was first adopted.¹⁹⁷

Some commenters stated that having ticks that are too small would result in queue jumping¹⁹⁸ and decreased depth.¹⁹⁹ In the Regulation NMS Adopting Release, the Commission

¹⁹⁴ See Letter from Matthew Gayvin Mutman dated Mar. 7, 2023. The commenter suggested a uniform \$0.001 minimum pricing increment for all NMS stocks. Comments related to the level of minimum pricing increment are addressed in the next section.

¹⁹⁵ See infra note 994 defining pennyning. See also infra section VII.D.1 for additional discussion of this topic.

¹⁹⁶ See, e.g., Form Letter Type G Nasdaq Letter I; MFA Letter; Letter from Douglas Friedman, General Counsel, Tradeweb Markets Inc., dated Mar. 30, 2023 ("Tradeweb Markets Letter"); Virtu Letter II; State Street Letter; RBC Letter; Invesco Letter; ICI Letter I; Cboe Letter II; SIFMA Letter II; Vanguard Letter; JPMorgan Letter; Hudson River Letter; T. Rowe Price Letter at 4; Goldman Sachs Letter; Fidelity Letter; Citadel Letter I; Robinhood Letter; GTS Letter; BlackRock Letter; Citigroup Letter; Fidelity Letter at 11; Themis Letter at 3; and Tastytrade Letter at 20.

¹⁹⁷ See Regulation NMS Adopting Release, supra note 4, at 37551.

¹⁹⁸ See, e.g., MFA Letter at 11, State Street Letter at 3, and RBC Letter at 3,

¹⁹⁹ See, e.g., Nasdaq Letter I at 13; MFA Letter at 11, Virtu Letter II at 15, State Street Letter at 3, and RBC Letter at 3.

discussed concerns related to stepping ahead of displayed quotations with orders priced in economically insignificant increments (i.e., to gain execution priority) which can deter the display of aggressively-priced limit orders that would narrow the spread.²⁰⁰ In light of these comments, amended Rule 612 has been simplified compared to what was proposed. Thus, the Commission is only adding the \$0.005 minimum pricing increment for quotes and orders for those NMS stocks that have a TWAQS of \$0.015 or less. Because the \$0.005 minimum pricing increment is based on the TWAQs of the NMS stock, the \$0.005 minimum pricing increment, relative to the spread, will be economically significant for these stocks.²⁰¹

Some commenters stated that smaller tick sizes would cause flickering quotations.²⁰² In the Regulation NMS Adopting Release, the Commission considered issues related to quote flickering.²⁰³ The Commission stated that quote flickering can result in broker-dealers having difficulties in satisfying their best execution obligations and other regulatory responsibilities.²⁰⁴ Because computer algorithms and ultra-fast connections dominate today's trading and quoting activities such concerns are not as acute or prevalent as they were at the time of the adoption of Rule 612.²⁰⁵ Today's quotations are calculated and displayed in microseconds, which is significantly faster than in 2005 and while flickering quotations can exist today, computer systems are much better able to process them such that they should not cause compliance

²⁰⁰ See Regulation NMS Adopting Release, supra note 4, at 37551.

²⁰¹ See infra section VII.D.1.b.ii and notes 1300-1303 and accompanying text.

²⁰² See, e.g., MFA Letter at 11, State Street Letter at 3, RBC Letter at 3, and Invesco Letter at 3.

²⁰³ See Regulation NMS Adopting Release, supra note 4, at 37551.

²⁰⁴ Id. at 37552.

²⁰⁵ See MDI Adopting Release, supra note 10, for a discussion about market data latencies. Flickering quotations is more of a concern when there is quote latency, in other words, when the displayed quotations do not reflect the actual quotations. For example, when the quote is being updated faster than the quote can be displayed, the price discovery mechanism may not be benefitted.

difficulties or investor confusion.²⁰⁶ Accordingly, because of technological advancements, today's market structure, compared to 2005, can more readily handle rapid changes to a trading center's best bid or offer. Further, the concerns about the potential for flickering quotes should be mitigated to some extent because the amendments do not include the smaller proposed increments (*i.e.*, \$0.001 and \$0.002) and are designed to have fewer ticks between the spread which will lessen the potential price changes between the spread.

Other commenters stated that the proposed minimum quoting increments of \$0.002 and \$0.001 were too small,²⁰⁷ would introduce too many intra-spread ticks,²⁰⁸ and could harm trading by substantially increasing fragmentation of liquidity.²⁰⁹ The Commission also considered the impact of sub-penny quoting on market depth,²¹⁰ *i.e.*, the number of shares available at the NBBO when it originally adopted quoting increments.²¹¹ Decreased depth could lead to increased transaction costs and fragmentation.²¹² Adopting only one additional minimum quoting increment instead of the proposed four-tier approach, should help address commenters' concerns with respect to fragmented liquidity²¹³ because there will be fewer price levels at which liquidity aggregates, which will result in less fragmentation. The modified amendment of Rule 612 does

²⁰⁶ See Regulation NMS Adopting Release, *supra* note 4, at 37553 - 37554 (discussing the concerns with flickering quotes when Rule 612 was adopted and acknowledging that the market could evolve).

²⁰⁷ See, *e.g.*, SIFMA Letter II at 33; Vanguard Letter at 5; Schwab Letter II at 35; Fidelity Letter at 11; JPMorgan Letter at 4; UBS Letter at 12; Citigroup Letter at 4; and Harris Letter at 7.

²⁰⁸ See, *e.g.*, Pragma Letter, Robinhood Letter at 40; IEX Letter I at 9; and Angel Letter at 6. The adopted \$0.005 minimum pricing increment will provide for at least three ticks intra-spread. See *infra* section VII.D.1.

²⁰⁹ See, *e.g.*, Interactive Brokers Letter at 4; Virtu Letter II at 4; and Themis Letter at 3.

²¹⁰ See *infra* section VII.D.1.b.

²¹¹ See Regulation NMS Adopting Release, *supra* note 4, at 37552.

²¹² See Regulation NMS Adopting Release, *supra* note 4, at 37552.

²¹³ See Citadel Letter I at 7. See *also* Virtu Letter II at 2 and 6-7.

not include the proposed smaller minimum pricing increments for quotes and orders of \$0.001 and \$0.002, and thus commenters' concerns related to those increments (e.g., decreased depth at the NBBO) are not applicable.²¹⁴ As discussed, the Commission has determined to take an incremental approach in amending Rule 612 by only adding a \$0.005 minimum pricing increment for those NMS stocks that are constrained by the preexisting, uniform minimum pricing increment based on an objective standard that is designed to have fewer ticks between the spread than the proposal.²¹⁵ As adopted, those NMS stocks that are assigned the \$0.005 minimum pricing increment will result in three ticks intra-spread, which falls in the middle of the 2 to 4 ticks intra-spread suggested as potentially optimal by many commenters,²¹⁶ Finally, the Commission addresses its primary concern of relieving the constraint related to the \$0.01 increment for certain NMS stocks by only adding the \$0.005 minimum pricing increment and not adding minimum pricing increments of \$0.002 and \$0.001. The \$0.005 minimum pricing increment for constrained NMS stocks will allow these stocks to quote more naturally and efficiently, and thereby reduce transaction costs for investors without the concerns that would attach if the minimum pricing increments were smaller.

3. Comments on the Number of Proposed Increments

Some commenters supported reducing the minimum pricing increment for quotes and orders to address those NMS stocks that are tick-constrained, but overall did not support the proposal's four minimum quoting increments.²¹⁷ Many commenters stated that the proposed

²¹⁴ See infra section VII.D.1.b.i.

²¹⁵ See infra section III.C.6.

²¹⁶ See infra note 1299 and accompanying text.

²¹⁷ See, e.g., SIFMA Letter II at 34; AIMA Letter at 2; STA Letter at 6-7; Citadel Letter I at 30; Citigroup Letter at 4; Dimensional Letter at 2; BlackRock Letter at 3; Public Pension Letters dated Mar. 31, 2023;

quoting increments were too numerous.²¹⁸ Instead, a number of commenters recommended that the Commission adopt a modified, simpler amendment to Rule 612 and suggested only adopting one additional minimum quoting increment of \$0.005 for tick-constrained NMS stocks.²¹⁹ One commenter said that “reducing the tick size to one-half cent for stocks with narrower spreads will address the current market need.”²²⁰ Commenters opposed the proposed four minimum quoting increments based on complexity for market participants to program into their systems these increments,²²¹ potential increased costs for investors,²²² and potential investor confusion with respect to minimum pricing increments that could change periodically as proposed.²²³ Another commenter stated that the four-tier proposal would favor “high-frequency traders who have a long history of leveraging complexity to their advantage and to the detriment of ordinary investors.”²²⁴ One commenter stated that the proposed variable minimum pricing increments “as

MMI Letter at 3; Brandes Letter at 1; Schwab Letter II at 35-36; Invesco Letter at 3; B. Riley Letter at 1; JPMorgan Letter at 4; Cambridge Letter at 6; and Tastytrade Letter at 18.

²¹⁸ See, e.g., MFA Letter at 12; Capital Group Letter at 3; ICI Letter I ; Angel Letter at 6 ; Vanguard Letter at 5; and Meuser et al. Letter at 1.

²¹⁹ See id. See also Nasdaq Letter I; MFA Letter; MEMX Letter; Capital Group Letter; ICI Letter I; Citadel Letter I; Citigroup Letter at 4; BlackRock Letter; Apex Letter; Ontario Teachers et al. Letter at 2; Citigroup Letter; GTS Letter; ICI Letter I; Invesco Letter; Robinhood Letter; SIFMA Letter II; STA Letter; UBS Letter; Vanguard Letter; TradeStation Letter at 6; Cboe Letter; IEX Letter; Nasdaq Letter I; and NYSE Letter I; Brandes Letter at 2; Invesco Letter at 2; Fidelity Letter at 14; Themis Letter at 6; B. Riley Letter at 1; JPMorgan Letter at 4; Morgan Stanley Letter at 4; State Street Letter at 3; Dimensional Letter at 2; BMO Capital Letter at 2; and Meuser et al. Letter at 1.

²²⁰ See ASA Letter at 5. See also TradeStation Letter at 6.

²²¹ See, e.g., CTA/UTP Letter dated March 29, 2023; Nasdaq Letter I; State Street Global Letter; RBC Letter; ICI Letter I; Vanguard Letter; Cboe Letter II; SIFMA Letter II; Fidelity Letter; Brandes Letter at 2; Robinhood Letter at 20; Morgan Stanley Letter at 4; and Meuser et al. Letter at 2.

²²² See, e.g., Dimensional Letter at 2.

²²³ See, e.g., Tastytrade Letter at 5, 18; SIFMA Letter II at 7; Morgan Stanley Letter at 3, 4; Fidelity Letter at 13; SIFMA Letter II at 34; Better Markets Letter I at 14; Robinhood Letter at 20; Citadel Letter I at 8; and STA Letter at 5.

²²⁴ See Better Markets Letter II at 4. See also Fidelity Letter at 12; Themis Letter at 6; Ontario Teacher et al. Letter at 2; and Harris Letter at 7.

small as \$0.001 goes well beyond what is necessary, and would also be cost prohibitive and complicated to implement.”²²⁵ One commenter questioned the impact of smaller increments on Rule 611 of Regulation NMS and recommended that if the Commission “proceed[ed] with their sub-penny quoting proposal. . . .”, it should consider amending Rule 611 to include all displayed depth of book quotes.²²⁶

After considering the comments and analyzing data,²²⁷ the Commission is amending Rule 612 to only add one new minimum pricing increment of \$0.005 for those NMS stocks that have a TWAQS of \$0.015 or less, rather than also adopting the additional two \$0.002 and \$0.001 pricing increments as proposed. The Commission’s basis for the new minimum pricing increment of \$0.005 is rooted by the current midpoint increment when the NBBO is at its narrowest (or smallest) spread. The midpoint increment of the current \$0.01 minimum quoting spread is calculated as (NBB plus NBO) divided by 2, and when the spread is at its narrowest, the midpoint increment is equal to \$0.005. For example, if the NBB is 10.01 and the NBO is 10.02, the midpoint would be 10.015 $((10.01+10.02)/2)=10.015$). Further, the new minimum quoting increment is at a price level familiar to all market participants and is already programmed into many computer systems. This modified approach addresses the concerns raised by commenters related to the proposed \$0.002 and \$0.001 minimum pricing increments. The adopted amendments also address commenters’ concerns about complexity and potentially

²²⁵ See TradeStation Letter at 6.

²²⁶ See Themis Letter at 5. As discussed, the Commission is adopting a modified amendment to Rule 612 to introduce only a \$0.005 minimum pricing increment for certain NMS stocks, not the smaller proposed increments of \$0.002 and \$0.001. Therefore, the commenter’s recommendation is no longer germane because without the proposed smaller \$0.002 and \$0.001 increments, the liquidity would not be as dispersed throughout the depth of the book which would not necessitate protection of the full depth of the book.

²²⁷ See infra section VII.D.1.

advantaging certain types of market participants by reducing the number of new increments and the universe of NMS stocks that may be eligible for a smaller minimum pricing increment. The adopted \$0.005 minimum pricing increment for those NMS stocks that have a TWAQS of \$0.015 will address the immediate concerns about the constraints that have developed in the national market system as a result of preexisting Rule 612.

4. Comments on Small- and Mid-Sized Stocks

A few commenters stated that the proposal to reduce minimum pricing increments did not consider the impact on small and mid-sized stocks.²²⁸ One commenter opposed the Regulation NMS Proposal because of concerns that it did not “address the needs and possible unintended consequences for small and mid-sized stocks” and that the Commission should “not take any action until such time as a pilot has been launched and its effects studied and verified by a committee of market participants and academics.”²²⁹ Another commenter stated that the proposed tick sizes were “too granular” for small to mid-sized stocks and would result in fewer liquidity providers.²³⁰

The assignment of the smaller minimum pricing increment is not based on market capitalization because the economics of being tick-constrained do not depend on market capitalization. Rather, whether a stock is experiencing constraint depends on its spread. In other words, since a stock’s spread relative to the tick size does not depend on whether it has a small or mid-sized market capitalization, such a stock could still trade with a quoted spread constrained by \$0.01 minimum pricing increment. With respect to implementing a pilot program to assess the

²²⁸ See BIO Letter at 1-2, 3 and STA Letter at 5.

²²⁹ See BIO Letter at 1-2, 3.

²³⁰ See STA Letter at 5.

needs and potential consequences of the proposal for small and mid-sized stocks, the Commission previously conducted a tick size pilot program for small- and mid-sized stocks to assess the impact of wider minimum quoting and trading increments.²³¹ The Commission analyzed data from that pilot program for purposes of the amendments.²³² Another pilot program is not necessary because the Commission and market participants have demonstrated with data the issues related to tick constraints that have increased since the preexisting rule was adopted.²³³ Further, the modified amendment will not introduce increments that are “too granular” for any NMS stock; only those NMS stocks that have a TWAQS of \$0.015 or less will be assigned the new \$0.005 increment, or three ticks or fewer within the spread. These NMS stocks are constrained by the preexisting increment and the amendment will alleviate this regulatory constraint to allow competitive forces of supply and demand to better establish bid and ask prices.²³⁴

5. Comments on Market Resiliency

A few commenters raised concerns related to market resiliency risks.²³⁵ The commenter stated that “[b]ecause the Commission’s proposal would increase the number of ticks inside the weighted average spread for many stocks, we could expect a significant increase in message traffic that would result from the Commission’s proposal.”²³⁶ The commenter asked the

²³¹ See Proposing Release, supra note 11, at 80272-73 for a discussion of the tick size pilot program. See also Tick Sizes and Market Quality: Revisiting the Tick Size Pilot by Yashar H. Barardehi, Peter Dixon, Qiyu Liu, and Ariel Lohr, available at https://www.sec.gov/dera/staff-papers/working-papers/dera_wp_tick-sizes-and-market-qualityrevisiting-tick-size-pilot.

²³² See infra section VII.D.1.

²³³ See infra section VII.D.1.b.ii.

²³⁴ See also infra section VII.D.1.b.i and VII.B.2 for additional discussion.

²³⁵ See, e.g., Letter from Howard Meyerson, Managing Director, Financial Information Forum, dated Mar. 31, 2023 (“FIF Letter”) at 6; and Goldman Sachs Letter at 8.

²³⁶ See FIF Letter at 7.